

Program: Diploma in Electrical Engineering& Electrical Vehicle Technology	
Course Code: 6411A	Course Title: Vehicle Power Train control Systems
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4(L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand general aspects of Electric and Hybrid Vehicles
- To understand the control system basics of EV's
- To understand basic design considerations of EV's
- To understand the basic concepts of braking systems and sensors

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic knowledge about storage batteries	2411	Introduction to Electric Vehicles &Elementary Theory	2
Basic Knowledge of batteries and charging systems	3411	Electric Vehicle Batteries & Charging Systems	3

Course outcomes:

On Completion of this course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Identify the various control systems & drive trains in EV & HEV	15	Understanding
CO2	Outline the basic design considerations of EV	15	Understanding
CO3	Outline the features of digital engine control systems	15	Applying
CO4	Power Electronics in HEV's	13	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the various control systems & drive trains in EV & HEV		
M1.01	Explain the control systems for EV's	4	Understanding
M1.02	Explain the split power devices for HEV	4	Understanding
M1.03	Identify various drive trains in Electric Vehicles and Hybrid Electric Vehicles	4	Understanding
M1.04	Identify the braking systems in EV	3	Understanding
Contents: Control systems for EV - Sizing the drive system - Sizing the propulsion motor, sizing the power electronics - Selecting the energy storage technology - Communications - supporting subsystems. Power split devices for Hybrid Vehicles. Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In Wheel Drives. Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.			
CO2	Outline the basic design considerations of EV		
M2.01	Describe the design requirements for electric vehicles	4	Understanding

M2.02	Summarize the dynamics and performance of electric vehicles during acceleration, rolling resistance ,aerodynamic drag ,vehicle moving up and down a hill	4	Understanding
M2.03	Explain the layout of power train control systems	4	Understanding
M2.04	Describe the components and modules of power train	3	Understanding
	Series Test I	1	
Contents: Design requirement for electric vehicles- Range, maximum velocity, acceleration, power Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive Effort and Transmission. Requirement, Transmission efficiency, energy consumption. Power train control system- Layout and various components of power train – ECU,PCM module , BCM module, BMS			
CO3	Outline the features of digital engine control systems		
M3.01	Illustrate the features of digital power train control systems	6	Understanding
M3.02	Explain various modes for fuel control	6	Understanding
M3.03	Explain Electronic ignition control	3	Understanding
Contents: Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control.			
CO4	Power Electronics in HEV's		
M4.01	Outline the PHEV configuration	5	Understanding
M4.02	Outline the basic ideas of power electronic systems	5	Understanding
M4.03	With the help of block diagram explain various power converters	3	Understanding
	Series Test II	1	

Contents:

PHEV configuration , power control unit, power electronics converters used in a series HEV

Basic Principles of power electronics- block diagram of power converter – types of power electronic converters -DC-DC converters, DC-AC converters, AC–AC converters, Rectifiers - Design of HEV power electronic circuits – Thermal Management of power electronic systems in EV

Text/Reference:

T/R	BookTitle/Author
R1	Jack Erjavec, Jeff Arias, “Hybrid, Electric and Fuel-cell Vehicles”, Cengage Learning India Pvt Ltd. New Delhi.
R2	S. Onori, L. Serrao and G. Rizzoni, “Hybrid Electric Vehicles: Energy Management Strategies”, Springer, 2015
R3	T. Denton, “Electric and Hybrid Vehicles”, Routledge
R4	A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi (Ed.
R5	C. Mi, M. A. Masrur and D. W. Gao, “Hybrid Electric Vehicles: Principles and
R6	Dr. P.S Bimbhra , “Power Electronics”, Khanna Publishing House ,New Delhi (Ed .
R7	Muhammed H Rashid, “Power Electronics Devices , Circuits & Applications”
R8	Dr. B R Gupta , “ Power Electronics”, Katson books

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/108/103/108103009/
2	https://nptel.ac.in/courses/108/106/108106170
3	https://www.sciencedirect.com/science/article/pii/S2095756420300647
4	https://www.metisengineering.com
6	https://onlinecourses.nptel.ac.in/noc20_ee99/preview